

CARIS DNA + RNA: research findings

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Source: nine Caris data files (DNA/RNA FASTQ and BAM, VCF, expression CSV and analytical workbook), from the 13 July 2026 FFPE brain specimen. Bulk sequencing; no matched-normal file supplied.

MTAP / CDKN2A / CDKN2B deletion. The workbook calls all three genes Deleted; exported CNA values are 0.80, 1.30 and 0.90, respectively. These are not independently calibrated tumor copy counts. MTAP RNA remains detectable (4.88 TPM); bulk residual signal cannot resolve tumor-cell copy state. **Open:** segmented copy number, purity/ploidy and allele-specific analysis to distinguish homozygous loss from other explanations.

Sequence-supported coding alterations. **BAP1** c.168_178del11 / p.R57fs, **RASA1** c.747delG / p.S250fs and **APC** p.S2577* have exported DNA VAFs of 54.2%, 53.5% and 40.3%, with mutant RNA support. All 21 selected protein-altering export calls were checked in DNA and RNA. Chromosome 3p allelic imbalance warrants copy-number/LOH review but does not establish a second hit. APC sequence reconstruction places its stop late in the protein; do not infer the same effect as an early APC truncation.

Additional Hippo-pathway candidates found in the reads. **LATS1** GRCh38 chr6:149695191 T>TA, ENST00000543571.6 c.378dup, predicts p.(Asn127Ter). Alternate/reference fragment counts: DNA **23/16**, RNA **4/2**. **LATS2** chr13:20975259 C>T, ENST00000382592.5 c.2878G>A, predicts p.(Gly960Arg): DNA **79/38**, RNA **44/5**. Neither appears in the supplied variant tables. A second implementation reproduced these counts/consequences. LATS1 is a truncation hypothesis; LATS2 function is unknown. Both need normal-sample exclusion and orthogonal validation.

RNA expression starting points. MSLN **253.01**, KRT5 **378.89**, KRT6A **119.35**, CALB2 **41.64** and WT1 **61.45 TPM**; EPCAM **3.02**, CLDN4 **0.60**, KRT20 **0.52** and UPK2 **0.49 TPM**. These are bulk measurements, not tumor-cell protein levels or an independently validated lineage classifier. MSLN is a focused protein-validation lead.

Most developed antigen hypothesis: BAP1 / HLA-B*50:01. Exact mutant coding windows support **IEERKGLYL** (691 DNA / 241 RNA fragments) and overlapping **EERKGLYL** (693 / 245). MHCflurry 2.3.0 predicts B*50:01 binding at **304 / 364 nM** (ranks 0.78% / 0.85%); both also scored favorably in the 2.2.0 check. RNA HLA typing yielded A*02:01/A*26:01, B*50:01/B*52:01 and C*06:02/C*12:02, stable across two reference releases. These are overlapping candidates from one mutation. Translation, tumor specificity, tumor HLA retention, peptide presentation and T-cell recognition remain untested.

Completed broader searches; useful limits. Full-input exploratory fusion analysis (23.2 million RNA pairs) produced **no high-confidence actionable fusion** after junction/BAM review. Of 143 selected alternative splice junctions, **141 match normal GTEx coordinates**. Broader indel screening retained 32 RNA-supported candidates; 26 match public population alleles. Bulk TCR reconstruction yielded only **one conservatively retained partial beta CDR3, (CASSLIGGDAFF)**, with no alpha-beta pairing or tumor-reactivity evidence. These bounded searches are not comprehensive negative clinical tests.

Highest-value extensions. Prioritize matched-normal somatic classification; MTAP/BAP1 allele-specific copy-number analysis; LATS1/LATS2 validation; BAP1 antigen-presentation testing; and genome-aware fusion/structural-variant analysis. Provisional HLA-II calls are available; class-II peptide scoring was not completed because public numerical controls failed. Methods, candidate tables, controls and audit outputs are retained for reproducibility.

Fragment counts collapse read pairs by name; they are not UMI-confirmed independent molecules. Research summary of the Caris data only; no pathology, imaging or treatment records used.